

Practical 2

2. Study of Network-Related Commands (Linux)

- a. **Hostname** : Shows computer's hostname.

```
sony@ubuntu-desktop: ~  
sony@ubuntu-desktop:~$ hostname  
ubuntu-desktop  
sony@ubuntu-desktop:~$
```

- b. **ip(earlier used as ipconfig)** : Used to view and configure network interfaces

```
sonworks@sonworks:~$ ip  
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }  
       ip [-force] [-batch filename]  
where  OBJECT := { address | addrlabel | fou | help | ila | loam | l2tp | link |  
                 macsec | maddress | monitor | mptcp | mroute | mrule |  
                 neighbor | neighbour | netconf | netns | nexthop | ntable |  
                 ntbl | route | rule | sr | tap | tcpmetrics |  
                 token | tunnel | tuntap | vrf | xfrm }  
OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |  
             -h[uman-readable] | -iec | -j[son] | -p[retty] |  
             -f[amily] { inet | inet6 | mpls | bridge | link } |  
             -4 | -6 | -M | -B | -O |  
             -l[oops] { maxnum-addr-flush-attempts } | -br[ief] |  
             -o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename]  
|  
-rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |  
-c[olor]]
```

```
chris@ubuntu-GT70: ~  
chris@ubuntu-GT70:~$ ping -c 5 google.com  
PING google.com (173.194.33.0) 56(84) bytes of data.  
64 bytes from sea09s01-in-f0.1e100.net (173.194.33.0): icmp_seq=1 ttl=58 time=17.9 ms  
64 bytes from sea09s01-in-f0.1e100.net (173.194.33.0): icmp_seq=2 ttl=58 time=42.1 ms  
64 bytes from sea09s01-in-f0.1e100.net (173.194.33.0): icmp_seq=3 ttl=58 time=16.0 ms  
64 bytes from sea09s01-in-f0.1e100.net (173.194.33.0): icmp_seq=4 ttl=58 time=27.0 ms  
64 bytes from sea09s01-in-f0.1e100.net (173.194.33.0): icmp_seq=5 ttl=58 time=10.9 ms  
  
--- google.com ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4005ms  
rtt min/avg/max/mdev = 10.909/22.817/42.149/10.991 ms
```

- c. **ping** : Used to test connectivity to another host.
- Shows:
 - Packet loss
 - Round-trip time
 - Whether network is up or down

```
kuts@FOSSLinux:~$ traceroute google.com  
traceroute to google.com (172.217.163.46), 30 hops max, 60 byte packets  
 1 DESKTOP-GCLV9EU.mshome.net (172.17.208.17)  0.223 ms  0.214 ms  0.209 ms  
 2 192.168.0.1 (192.168.0.1)  0.995 ms  1.466 ms  1.464 ms  
 3 172.19.140.1 (172.19.140.1)  2.411 ms  2.397 ms  2.169 ms  
 4 172.30.5.242 (172.30.5.242)  4.155 ms  4.669 ms  4.378 ms  
 5 172.30.6.65 (172.30.6.65)  3.459 ms  3.447 ms  3.650 ms  
 6 10.241.1.6 (10.241.1.6)  15.381 ms  14.615 ms  14.211 ms  
 7 10.240.254.130 (10.240.254.130)  8.862 ms  1.882 ms  2.183 ms  
 8 10.240.254.1 (10.240.254.1)  8.416 ms  2.917 ms  2.642 ms  
 9 10.241.1.1 (10.241.1.1)  3.129 ms  2.065 ms  3.346 ms  
10 172.30.1.57 (172.30.1.57)  1.823 ms  3.538 ms  2.389 ms  
11 ns0.wishnet.in (113.21.64.1)  4.348 ms  4.105 ms  4.335 ms  
12 ns0.wishnet.in (223.223.158.197)  45.832 ms  45.618 ms  45.530 ms  
13 74.125.242.145 (74.125.242.145)  46.257 ms  44.058 ms  74.125.242.129 (74.125.242.129)  45.265 ms  
14 216.239.42.215 (216.239.42.215)  44.814 ms  44.813 ms  45.014 ms  
15 maa05s01-in-f14.1e100.net (172.217.163.46)  44.061 ms  44.497 ms  44.276 ms
```

- d. **Traceroute** :
- Shows the path packets take to reach a destination.
 - Helps identify where network delay or failure occurs.
 - Displays each hop's IP and response time.

- e. **nslookup** : used to query DNS servers to find the IP address of a domain or the domain name of an IP.

```
student@Comp9:~$ nslookup google.com  
Server:         127.0.0.53  
Address:        127.0.0.53#53  
  
Non-authoritative answer:  
Name:   google.com  
Address: 172.217.167.174  
Name:   google.com  
Address: 2404:6800:4009:810::200e
```

```
tuts@FossLinux: ~  
File Edit View Search Terminal Help  
tuts@FossLinux:~$ netstat -a | more  
Active Internet connections (servers and established)  
Proto Recv-Q Send-Q Local Address           Foreign Address         St  
ate  
tcp        0      0 localhost:domain        0.0.0.0:*               LI  
STEN  
tcp        0      0 localhost:ipp            0.0.0.0:*               LI  
STEN  
tcp6       0      0 ip6-localhost:3350      [::]:*                  LI  
STEN  
tcp6       0      0 ip6-localhost:ipp       [::]:*                  LI  
STEN  
udp        0      0 0.0.0.0:53188           0.0.0.0:*                 
udp       32256      0 localhost:domain        0.0.0.0:*                 
udp        0      0 localhost:ntp            0.0.0.0:*                 
udp        0      0 0.0.0.0:ntp             0.0.0.0:*               
```

f. **netstat(now used as ss) :**

- Displays **network connections, routing tables, ports, etc.**
- Here, **netstat -a :** Show all active connections